

A Study on (S)-Ibuprofen: From Conformer Generation to Molecular Docking

Project Report

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Introduction

The advancement of computational techniques has significantly transformed the drug discovery as we know it, providing powerful tools to researchers to explore the structural and functional properties of potential therapeutic compounds before they reach the laboratory. These *in silico* methods, which include conformational sampling, quantum mechanical calculations, and molecular docking, allow for a detailed molecular level understanding of drug behaviours, offering a cost effective and time efficient complement to experimental procedures (*Sliwoski et. al., 2013*). This project applied some of the computational methodologies mentioned to the molecule (S)-ibuprofen, in hopes of investigating its conformational diversity, electronic properties, and interactions with a biological target.

(S)-Ibuprofen is the pharmacologically active enantiomer of ibuprofen, a well known nonsteroidal anti inflammatory drug (NSAID) widely used for its analgesic, antipyretic, and anti inflammatory effects. Its therapeutic activity stems from the inhibition of cyclooxygenase (COX) enzymes, which catalyse the biosynthesis of pro inflammatory prostaglandins (*Gažová et. al., 2021*). Notably, the (S)-enantiomer is substantially more potent than its (R)-counterpart, a distinction that highlights the critical role of stereochemistry in drug efficacy. Structurally, (S)-ibuprofen features a chiral centre adjacent to a carboxylic acid and a bulky isobutyl group, attributes that contribute to its conformational flexibility and the complexity of its interactions with protein targets (*Orlando et. al., 2014*).

In this project, a complete computational workflow was implemented to model the behaviour of (S)-ibuprofen. The first phase involved the generation of multiple conformers to capture the molecule's structural flexibility. This was achieved through two distinct approaches: a genetic algorithm (GA) method, and a systematic method (Confab). The goal of this step was to ensure a comprehensive sampling of conformational space, which is vital for identifying the most bioactive geometries (*O'Boyle et. al., 2011*).

Following the conformer generation step, quantum chemical calculations were employed to refine and analyse the electronic properties of the molecule. Initially, semi empirical (PM7) calculations were performed to estimate the heats of formation of the generated conformers in an aqueous environment to identify the most energetically stable structures under each condition. These structures were then subjected to more accurate geometry optimization (*Hehre, 2021*). These quantum mechanical analyses provided insight into the electronic stability and optimized geometries of (S)-ibuprofen, ensuring that the most realistic molecular models would be used in subsequent docking simulations.

The final phase of the study consisted of molecular docking, in which the DFT refined structure of (S)-ibuprofen were docked into a biologically relevant protein target (3N8Y). The docking protocol involved defining the ligand's rotatable bonds, configuring a grid box around the protein's active site, and running docking simulations to predict the most favourable binding modes and associated energies. Visualization and interpretation of these binding poses allowed for the examination of key interactions such as hydrogen bonds and hydrophobic contacts between the ligand and the protein (*Fan et. al., 2019*).

Overall, this project integrates a robust set of computational techniques to explore the structural and functional characteristics of (S)-ibuprofen. By combining conformer

generation, electronic structure analysis, and molecular docking, it provides a detailed model of how this molecule behaves in a biological context, offering valuable insights that could support future drug design efforts and the development of improved NSAID analogues.

Methods

The computational investigation of (S)-ibuprofen was conducted through a multi-step workflow, in which, each phase was executed using widely used tools and protocols in computational chemistry, ensuring the reproducibility and accuracy of the results.

1. Conformational Analysis

The first stage involved generating conformational ensembles of (S)-ibuprofen to capture its intrinsic structural flexibility. This step is essential, as biologically active conformations may not always correspond to the molecule's lowest energy geometry. The conformer generation was performed using Open Babel, an open source cheminformatics toolkit capable of generating and manipulating 3D molecular structures. Two distinct methods were applied: the genetic algorithm (GA) approach and the Confab method. In the GA based generation, thirty unique conformers were created by optimizing Root Mean Square Deviation (RMSD) between structures, emphasizing diversity while maintaining low energy configurations. On the other hand, the Confab method systematically explored torsional space to generate an extensive set of conformers, targeting broad coverage of low energy geometries. Both sets of conformers were visualized using PYMOL, which enabled qualitative analysis and spatial comparison of molecular geometries through a grid style layout.

2. Electronic Structure Calculations

After generating our conformers, the electronic structure of the molecule was refined through quantum mechanical calculations, which provided more accurate information on the molecule's geometry and thermodynamic stability. The initial screening of conformers was carried out using the semi empirical PM7 method implemented using MOPAC. For this purpose, the three lowest energy conformers from each generation method were converted to .MOL format and prepared for calculation using AvogadroX11, which offers integrated support for input generation. Calculations were performed in an aqueous (water) environment to simulate physiological conditions (*Hirschfelder, 1941*). The output files were parsed to extract the final heat of formation values, which served as the primary metric for selecting the most stable conformers in each medium .

3. Quantum Mechanical Optimization

The most stable conformers identified through PM7 were subjected to a more rigorous optimization using the Density Functional Theory (DFT) at the B3LYP/6-31G(d, p) level of theory. These calculations were conducted using GAMESS, a quantum chemistry package designed for high level electronic structure modelling (*Lee, 2021*). Before launching DFT calculations, the optimized PM7 geometries were exported and opened in: wxmacmolplt, where the input files were created. Important computational parameters

were adjusted, including setting the basis set to “6-31G”, selecting “optimization” as the run type, and enabling sufficient memory allocation. Geometry optimization was carried out in water, with solvation effects modelled using the Polarizable Continuum Model (PCPM), which was enabled by adding the appropriate “SOLVNT=water” keyword to the input file. Each calculation was executed in a multi core environment, and successful convergence was verified by searching the output files for the term “EQUILIBRIUM GEOMETRY LOCATED”. In some cases, we faced a non convergence, in those cases continuation files were generated to restart the optimization from the final geometry of the previous run.

4. Molecular Docking

Once the molecular structure of (S)-ibuprofen had been fully optimized, it was prepared for molecular docking using Auto Dock Vina. This phase aimed to simulate the interaction between the drug and our defined protein target Aspirin Acetylated Cyclooxygenase-1 (3N8Y, in complex with Diclofenac), mimicking a real biological binding event. The target protein was retrieved from the Protein Data Bank (PDB) and pre processed using the program Visual Molecular Dynamics (VMD). All of the non essential molecules, such as water, were removed, and only one chain of the protein structure was kept. The cleaned structure was then saved in PDB format for further preparation. Ligand and receptor files were converted to PDBQT format using Auto Dock Tools (ADT), where polar hydrogens were added and rotatable bonds defined for the ligand. A grid box was configured around the active site based on the known binding region, with centre coordinates and box dimensions recorded to guide the docking simulation (Grid Points: X dimension = 126, Y dimension = 126, Z dimension = 126; Spacing(Å) = 0,475; Centre Grid Box: X = -36,654, Y = -51,733, Z = 2,08). The docking itself was executed via Auto Dock Vina using a configuration file containing the ligand and receptor paths, grid parameters, and search settings. The parameters included an exhaustiveness value, used to determine the thoroughness of the search, an energy range (kcal/mol) to capture alternative binding poses, and a maximum of output binding modes.

The results of the docking simulations were analysed visually using PYMOL, where the protein-ligand complexes were loaded and inspected. This visualization enabled the identification of potential hydrogen bonds, hydrophobic interactions, and the general fit of (S)-ibuprofen within the active site of the protein. These interactions provided insight into the binding mode and predicted affinity.

Results

The results from each phase produced intermediate data and visual evidence of the molecule's structural behaviour and interaction potential, ultimately leading to conclusions about its stability and binding characteristics.

The initial step involved generating conformers of (S)-ibuprofen using two complementary methods in Open Babel: the genetic algorithm (GA) and Confab. As expected, the GA approach quickly provided a diverse set of thirty low energy conformers, optimized for RMSD variation (Figure 1). These structures illustrated the rotational flexibility around the isobutyl and carboxylic moieties. In contrast, the Confab method produced a

more exhaustive conformational ensemble, systematically exploring torsional angles and yielding a dense population of structurally similar yet distinct conformers (Figure 2). The expected outcome was to visualize the full conformational space accessible to (S)-ibuprofen, identifying possible geometries that the molecule might adopt. The images generated through PYMOL (displayed through grid mode) confirmed that each conformation occupied a slightly different region in 3D space, with noticeable variations in their side chain orientation and overall molecular curvature, particularly in the orientation of the chiral centre and terminal methyl groups.

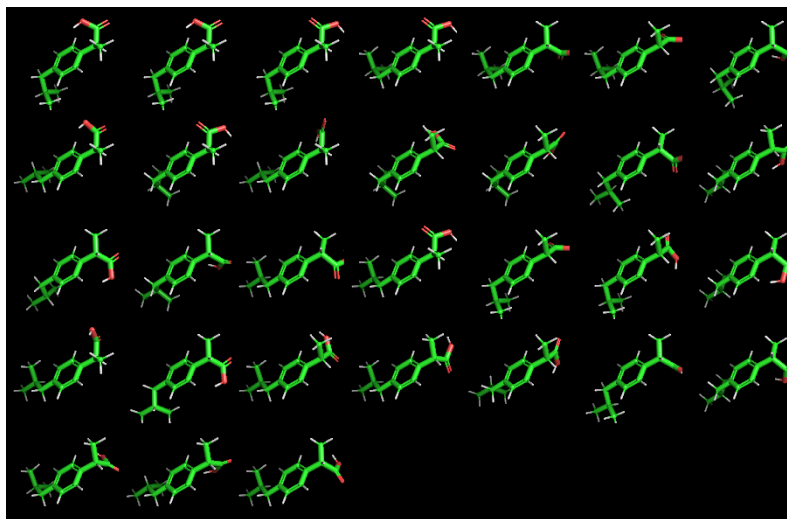


Fig. 1 – PYMOL grid visualization for the thirty different low energy generated conformers, obtained in Open Babel using the Genetic Algorithm (GA) complementary method optimized for RMSD variation.

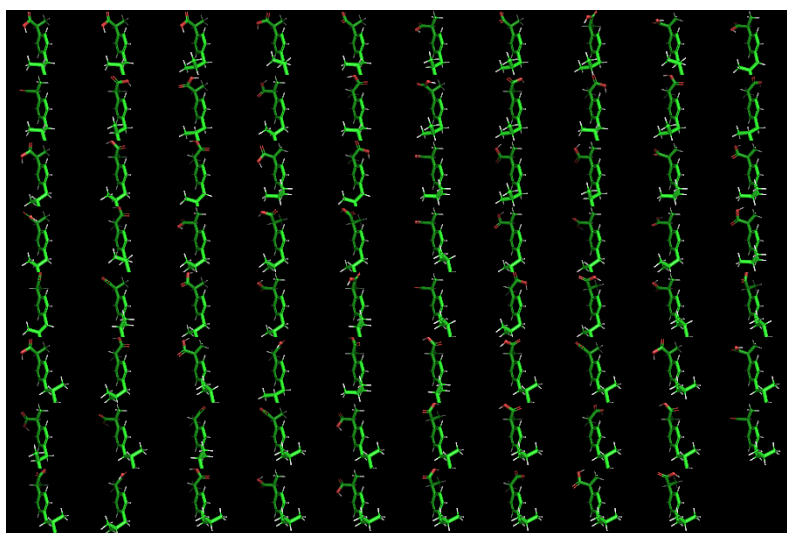


Fig. 2 – PYMOL grid visualization for the different low energy generated conformers, obtained in Open Babel using the Confab complementary method optimized for RMSD variation.

Following conformational sampling, electronic structure calculations were conducted using the PM7 semi empirical method for the six lowest energy conformers (the first three of each of the methods used – GA and Confab) and subsequently refined via DFT using the B3LYP/6-31G(d, p) level of theory. As expected, the PM7 calculations provided a rapid assessment of conformational energies, allowing for the identification of the most stable structures in an aqueous environment. After the calculations, we then chose the best conformer out of each of the three chosen from each method based on the ones that had the best value of “FINAL HEAT OF FORMATION”, in which the more negative the value the better (Table 1). These calculations confirmed that solvation has a stabilizing effect on the polar carboxylic acid group, with slight energy shifts observed in water simulations, with the best conformer for each method being the first one.

<i>Conformers</i>	<i>Final Heat of Formation:</i>
<i>GA_conf1.out</i>	- 108.52603 Kcal / mol
<i>GA_conf2.out</i>	- 108.17847 Kcal / mol
<i>GA_conf3.out</i>	- 105.14203 Kcal / mol
<i>Confab_conf1.out</i>	- 109.99542 Kcal / mol
<i>Confab_conf2.out</i>	- 109.50241 Kcal / mol
<i>Confab_conf3.out</i>	- 109.42841 Kcal / mol

Table 1 – Values for Final Heat of Formation of each of the three conformers chosen from the GA and Confab methods for the (S)-ibuprofen molecule

The DFT refinements yielded highly optimized structures, with convergence verified by the presence of “EQUILIBRIUM GEOMETRY LOCATED” in the output files. These geometries, when visualized, exhibited smoother bond angles and consistent electron distribution, reflecting the higher accuracy of the DFT method. Compared to their PM7 counterparts, the DFT optimized conformers had slightly different dihedral angles, indicating a more realistic representation of (S)-ibuprofen’s geometry in both states.

The final stage of the study was the molecular docking of the DFT optimized (S)-ibuprofen structure to the target protein retrieved from the Protein Data Bank. After preprocessing the protein to retain a single functional chain and defining the active site via a grid box, the ligand was prepared in PDBQT format, with torsions properly defined to reflect its flexibility. Auto Dock Vina simulations were run using high exhaustiveness settings to ensure thorough sampling of the conformational binding space. The docking output revealed several plausible binding poses (Figure 3), of which the top ranked (based on binding affinity in kcal/mol) were further analysed in PYMOL.



Fig. 3 – PYMOL representation of the docking results for several plausible binding poses of the (S)-ibuprofen structure (cyan blue) to the single retained functional chain of the Aspirin target protein (yellow).

The docking results aligned well with expectations for a COX inhibitor. The two best scoring poses positioned the carboxyl group of (S)-ibuprofen near the active site's hydrogen bond donors, suggesting potential polar interactions that could stabilize binding. The isobutyl side chain nestled into a hydrophobic pocket, replicating known interactions seen in crystal structures of COX-inhibitor complexes. The visual output for these two poses from PYMOL supported this interpretation: the ligand assumed a bent conformation that maximized both hydrogen bonding and van der Waals interactions. In several of the binding modes, the molecule adopted a shape highly reminiscent of the canonical NSAID binding posture, which further validated the quality of the docking results (Figure 4).

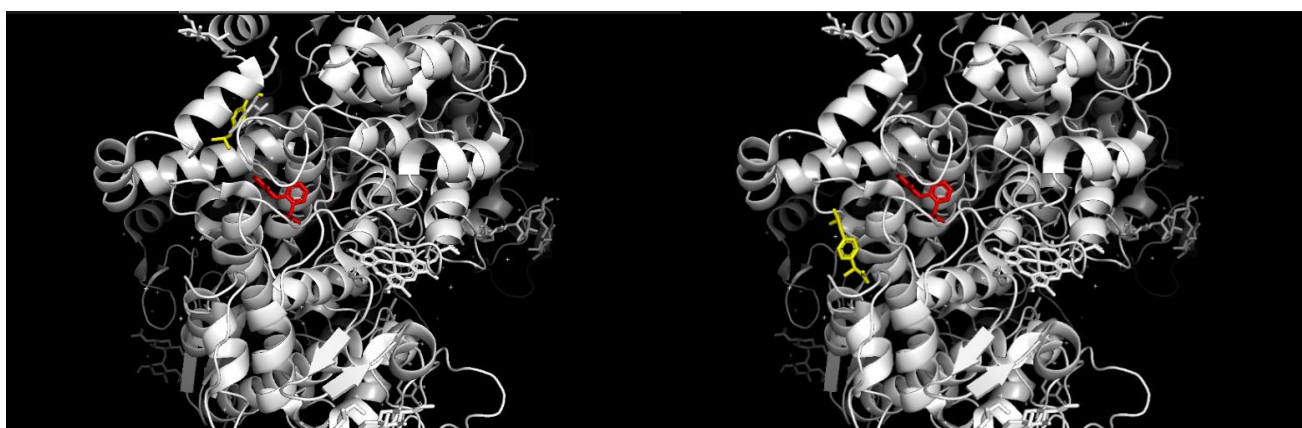


Fig. 4 – PYMOL visualization of the two best scoring binding positions (yellow) in contrast to the originally defined binding position in the binding pocket (red) in the Aspirin target protein (white).

Conclusion

Overall, the results obtained from each stage were consistent with the expected theoretical and pharmacological behaviour of (S)-ibuprofen. The conformer generation illustrated its flexibility, the electronic structure calculations identified energetically favourable conformations, and the docking simulations predicted realistic and pharmacologically relevant interactions with a protein target. These findings confirm the effectiveness of the computational pipeline and reinforce the reliability of the tools employed. Most importantly, they validate that the conformational and electronic properties of (S)-ibuprofen, as derived computationally, correlate well with its known bioactivity. This demonstrates that such an approach can be reliably used for other drug candidates in early stage drug discovery.

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